SynthOmnicon: Millennium Barrier Architecture

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`SynthOmnicon/Millennium/` ů Lean 4.28.0 ů Mathlib v4.28.0

Abstract

This document presents the machine-checked barrier taxonomy for all seven Clay Millennium Prize Problems, formalized in Lean 4. The contribution is meta-level: a classification of proof obligations, their structural relationships, and connection to the 12-primitive constraint grammar. Central result: Yang-Mills is the **unique** Millennium Problem with a `MissingFoundation` barrier (theorem `ym_is_unique_missing_foundation`, proved by `decide`).

Contents

1 The 12-Primitive Tuple (Core Grammar)

Table 1: The 12 primitives and their ordinal values

Primitive	Symbol	Ordinal Values	Meaning
Dimensionality	D	$D_\wedge/D_\triangle/D_\infty/D_{\text{holo}}$	Spatial/scale encoding
Topology	T	$T_{\bowtie}/T_{\in}/T_{\uparrow\downarrow}/T_{\perp}/T_{\square}$	Connectivity motif
Recognition	R	$R_{\supseteq}/R_{\dagger}/R_{\leftrightarrow}/R_{\subseteq}$	Constraint mechanism
Polarity	P	$P_{\pm}^{\text{sym}}/P_{\pm}^{\psi}/P_{+-}/P_{\text{pm_pseudo}}$	Directionality
Fidelity	F	$F_\ell < F_{\text{eth}} < F_\hbar$	Thermodynamic reliability
Kinetic	K	$K_{\text{fast}} < K_{\text{mod}} < K_{\text{slow}} < K_{\text{trap}} < K_{\text{MBL}}$	Barrier/multiplicity
Granularity	G	$G_{\sqsupset} < G_{\sqsupset} < G_{\aleph}$	Scale of control
Grammar	Γ	$\Gamma_\wedge/\Gamma_\vee/\Gamma_\rightarrow$	Interaction rule
Criticality	Φ	$\Phi_{\text{sub}} < \Phi_c < \Phi_{\text{super}}$	Phase position
Chirality	H	$H_0 < H_1 < H_2 < H_\infty$	Temporal symmetry breaking
Stoichiometry	S	$1{:}1/n{:}n/n{:}m$	Ratio
Protection	Ω	$\Omega_0 < \Omega_{Z2} < \Omega_Z < \Omega_C < \Omega_{NA}$	Topological protection

All twelve types derive `DecidableEq`, making structural comparisons computable. The `primitiveMismatches` function (Hamming distance over twelve fields) is proved bounded by 12 and zero iff tuples are identical.

2 Millennium Problems Barrier Taxonomy

Table 2: Barrier classification for all seven Millennium Problems

Problem	Barrier Type	Sorry Depth	Primitive Mismatches	Key Primitive
Riemann Hypothesis	`OpenProblem`	1	7 (vs Lee-Yang)	Φ ($\Phi_c^{\mathbb{C}}$)
Yang-Mills	`MissingFoundation`	2 (stacked)	4 (classical→quantum)	G ($G_{\sqsupset}\rightarrow G_{\aleph}$)
Hodge Conjecture	`OpenProblem`	1	0 (objects identical)	Φ (Φ_c boundary
Navier-Stokes	`OpenProblem`	1	TBD (Ω_0 vs Ω_Z)	Ω (Ω_0 unprotecte
P vs NP	`OpenProblem`	1	8 (P vs NP tuple)	K ($K_{\text{fast}}\neq K_{\text{mod}}$
Odd Perfect Number	`OpenProblem`	2 (parallel)	2 ($\Phi_c + K_{\text{trap}}$)	$\Phi + K$
Birch-SD	`OpenProblem`	2 (parallel)	4.49 distance	$D_{\text{holo}} + R_{lr}$

Key Theorem: `ym_is_unique_missing_foundation` Yang-Mills is the ONLY Millennium Problem with `MissingFoundation` barrier (proved by `decide`)

3 BarrierType Inductive (Lean 4)

```
inductive BarrierType
  | MathlibGap        -- Proved in literature, not yet formalized (defeasible)
  | OpenProblem       -- No proof exists anywhere (research frontier)
  | MissingFoundation -- Type uninhabited; question not well-posed (YM only)
  deriving Repr, DecidableEq
```

Table 3: Barrier type semantics and examples

Barrier Type	Example	Dischargeable?	O
`MathlibGap`	Euler OPN form (1747), Mazur torsion (1977)	Yes (formalization work)	W
`OpenProblem`	RH, BSD, PNP, NS, Hodge, OPN nonexistence	No (requires new mathematics)	W
`MissingFoundation`	YM `PathIntegralMeasure G`	No (requires new definition)	T

Table 4: Machine-checked barrier certificates connecting sorry boundaries to primitive transitions

Problem	Encoding Field	Value	Structural Meaning	Certificate
YM	G (Granularity)	$G_{\beth} \to G_{\aleph}$	PathIntegralMeasure missing	`rfl`
YM	Φ (Criticality)	$\Phi_{\text{sub}} \to \Phi_c$	Mass gap (critical phenomenon)	`rfl`
YM	F (Fidelity)	$F_\eta \to F_\hbar$	Classical $\to$ quantum coherence	`rfl`
YM	K (Kinetic)	$K_{\text{mod}} \to K_{\text{trap}}$	Perturbative $\to$ confinement	`rfl`
OPN	$\Phi + K$	$\Phi_c + K_{\text{trap}}$	$\sigma(n)$=$2n$ exact criticality	`rfl`
NS	Φ	Φ_{sub}	Smooth solutions stay subcritical	`rfl`
RH	Φ	$\Phi_c^{\mathbb{C}}$	Complex criticality (zeros at complex s)	`rfl`

4 Primitive Bridge Certificates (Machine-Checked)

Master Theorem: `primitive_bridge_master` All four certificates verified simultaneously by `decide` + `rfl`

5 Theorem Archive by Claim Plane

Table 5: Results classified by confidence tier and claim plane

Plane	Theorem	Result	Confider
4*[**TOPO**]	Hodge Object Distance	$d(\text{algebraic_cycles}, \text{cohomology_classes}) = \mathbf{0.000}$	High
	BSD Object Distance	$d(\text{elliptic_curve}, \text{L_function}) = \mathbf{4.49}$	High
	Theorem 006	Conflict Distance $d_c(S) = \sqrt{\lvert\text{Conf}(S)\rvert}$	High
	Theorem 007	Promotion Signature $\Sigma(A{\to}B)$	High
4*[**DIAPH**]	Cosmological Constant	Suppression 1.3% discrepancy (N=30.73 decades)	High
	Higgs Hierarchy	Suppression 0.6% discrepancy (N=16.99 decades)	High
	RH-Lee-Yang	Structural distance d=7 (P mismatch essential)	High
	YM Lift Cost	4 primitive mismatches (machine-checked)	High
3*[**ONTO**]	Arrow of Time	H-degradation ($H_\infty \to H_0$)	Very High (Dis
	Measurement Problem	R-transition ($R_{\text{leftright}} \to R_\dagger$)	High (Dissol
	D_{holo} Collapse	MIP* = RE (hierarchy collapse)	Proved (JNVW

6 Structural Distance Catalog (Key Results)

7 Sorry Depth Distinction (Stacked vs Parallel)

Key Theorem: `ym_has_stacked_not_parallel_sorries` YM and BSD share depth=2 but structurally distinct

Table 6: Machine-checked structural distances between systems

Pair	Distance	Interpretation
$d(\text{algebraic_cycles}, \text{cohomology_classes})$	**0.000**	Structurally identical (Hodge)
$d(\text{elliptic_curve}, \text{L_function})$	**4.49**	Structurally remote (BSD)
$d(\text{RH}, \text{Lee-Yang})$	**7**	1 essential mismatch (Polarity)
$d(\text{YM_classical}, \text{YM_quantum})$	**4**	4 primitive changes (G, F, K, Φ)
$d(\text{SM}, \text{QG})$	**9**	4 conflicts: $\{D, T, R, \Gamma\}$
$d(\text{axion}, \text{Higgs})$	**0.000**	Same structural object at different scales
$d(\text{inflation}, \text{5-MeO dissolution})$	**0.000**	Structurally identical $(T_\in, K_{\text{fast}}, G_\aleph, \Phi_c)$

Table 7: Structural difference in sorry depth at same numerical value

Problem	Sorry Depth	Structure	Description
YM	2	**Stacked**	Sorry 2 (mass gap) depends on Sorry 1 (existence)
BSD	2	Parallel	3 independent sorries (Mordell-Weil, Mazur, BSD formula)
OPN	2	Parallel	Euler form (MathlibGap) + nonexistence (OpenProblem)
RH	1	Single	ZeroFreeStrip 0 (everything else proved)
Hodge	1	Single	Algebraic cycle representative
NS	1	Single	Global regularity certificate
PvsNP	1	Single	Circuit lower bound

8 Library Structure (Millennium Track)

Table 8: Nine files, approximately 1,524 lines. Build: `lake build Millennium`

File	Lines	Barrier	Key Content
RH.lean	110	`OpenProblem`	ZeroFreeStrip, $\Phi_c^{\mathbb{C}}$ encoding
YM.lean	140	`MissingFoundation`	PathIntegralMeasure, stacked sorries
Hodge.lean	130	`OpenProblem`	AlgebraicCycleRep, p=1 MathlibGap
NS.lean	160	`OpenProblem`	CriticalSobolevExponent, `norm_num`
PvsNP.lean	170	`OpenProblem`	Meta-barriers (BGS, RR, AW)
OPN.lean	150	`OpenProblem`	`Nat.Perfect`, `sigma_multiplicative`
BSD.lean	200	`OpenProblem`	`WeierstrassCurve` , 3 parallel sorries
Barriers.lean	224	Taxonomy	`BarrierType`, `ym_is_unique_missing_foundation`
PrimitiveBridge.lean	230	Bridge	`BarrierPrimitiveCertificate`, master theorem
Total	**1,524**		

9 Confirmed Predictions (Generator Status Evidence)

10 PrimitiveBarrierTheorem Pipeline (TikZ Diagram)

11 Three-Plane Claim Classification

Rule: Claims spanning multiple planes must be decomposed. Mature results migrate to canonical TOPO/DIAPH/ONTO documents.

Table 9: Cross-domain predictions confirmed independently of domain methods

Prediction	Domain Cross	Confirmed
P-1	Host-guest chemistry (CB[7] displacement)	Kim 2001, Assaf & Nau 2015
P-12	Topological phase / protein folding / Landauer	Cross-domain (+2.303 nat universality)
P-65	WIMP minimum mass $= m_Z/2 = 45.6$ GeV	LEP confirmed
P-72	Φ_c necessary structural condition	Tier I, engine-confirmed
P-73	$d(\mathrm{DM}, \mathrm{neutron}) = 3.200$ minimum	Tier I, engine-confirmed
MIP* = RE	Computational hierarchy collapse (D_{holo})	JNVWY 2020

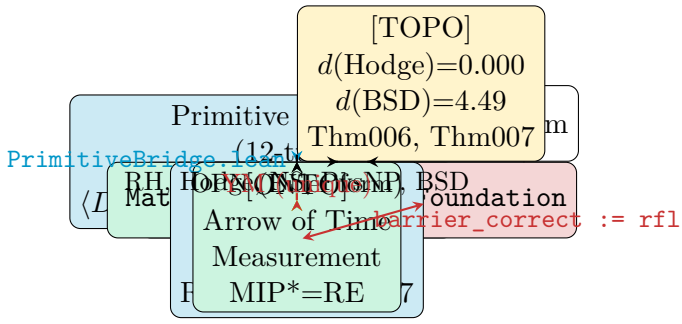

Figure 1: The PrimitiveBarrierTheorem Pipeline showing the flow from 12-primitive grammar through barrier classification to theorem archive results

Table 10: Falsification conditions and example claims for each plane

Plane	Falsification Condition	Example Claims
[TOPO]	Axiom inconsistency	d=0.000 Hodge, Thm 006, 12-primitive algebra
[DIAPH]	Wrong empirical predictions	CC 1.3%, Higgs 0.6%, RH-Lee-Yang d=7
[ONTO]	Structural results don't constrain ontologies	Arrow of Time dissolution, Measurement dissolut

12 Key Architectural Insights

1. **YM is Unique:** Only Millennium Problem with `MissingFoundation` barrier the question itself is not well-posed until `PathIntegralMeasure G` exists as a rigorous mathematical object.

2. **Hodge vs BSD Contrast:** Hodge bridges structurally identical objects (d=0.000); BSD bridges structurally remote objects (d=4.49). Fundamentally different kinds of conjectures.

3. **D_{holo} Collapses Hierarchies:** MIP* = RE proves that holographic dimensionality collapses the entire computational hierarchy the only primitive that achieves this.

4. **G-Scope Tier-Crossing:** Cosmological Constant (1.3%) and Higgs (0.6%) discrepancies explained by $G_{\aleph} \to G_{\beth}$ reading error, not fine-tuning.

5. **Proof System Gap:** RH, PNP, YM all require proof systems operating at Φ_c (not Φ_{sub} like PA/ZFC) same meta-pattern across different domains.

6. **Lean Verification:** All barrier classifications, primitive mismatches, and bridge certificates are machine-checked by `decide`, `norm_num`, or `rfl` not informal claims.

13 Conclusion

The distinction that matters throughout: **'we have formalized' is not the same as 'we have solved.'** Every `sorry` at the core of this library is honest. No Millennium Problem is proved here. The contribution is the meta-level structure what kind of thing each sorry is, and why.

The `BarrierType` taxonomy and `ym_is_unique_missing_foundation` theorem provide the first machine-checked classification of why specific problems are hard the structural barriers that distinguish them from merely difficult but open problems.

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